$$f(x)=\ln\left(x^4+20x^3+100\right)$$

$$f'(x)=\frac{1}{x^4+20x^3+100}\left(4x^3+60x^2\right)$$

$$f'(x)=\frac{4x^3+60x^2}{x^4+20x^3+100}$$

$$4x^3+60x^2=0$$

$$4x^2(x+15)=0$$

$$x=-15, x=0$$

increasing $\rightarrow (0,20)$